

Multiple regression

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One predictor is not enough

Holding something fixed

Is the model any good?

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It only knows what is on the record.
So what should it look at?



- 1,710 real home sales, Saratoga County, NY
- **price** – what it sold for. Median \$189,900.
- **bedrooms** – 2 to 5
- **livingArea** – interior square feet. Median 1,634.



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Obviously. More bedrooms, more house, more money.

But bigger houses also have more bedrooms



Predictor	Slope	R^2
bedrooms	\$53,048 per bedroom	0.172
livingArea	\$116 per sq ft	0.515

They are correlated: $r = 0.65$.

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So what happens if we use both at once?

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One predictor:

$$Y_i = \beta_0 + \beta_1(\text{bedrooms}) + \epsilon_i$$

Two predictors:

$$Y_i = \beta_0 + \beta_1(\text{bedrooms}) + \beta_2(\text{sqft}) + \epsilon_i$$



One predictor:

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Same idea. Still least squares. We just choose three numbers instead of two.

```
model <- lm(price ~ bedrooms + livingArea, data=homes)
summary(model)
```

```
Call:
lm(formula = price ~ bedrooms + livingArea, data = homes)
```

```
Residuals:
```

	Min	1Q	Median	3Q	Max
	-283919	-39664	-7421	28383	551848

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	28852.164	6954.957	4.148	3.51e-05 ***
bedrooms	-11925.646	2831.767	-4.211	2.67e-05 ***
livingArea	125.835	3.582	35.135	< 2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 68220 on 1707 degrees of freedom
```

```
Multiple R-squared:  0.5196, Adjusted R-squared:  0.519
```

```
F-statistic: 923.2 on 2 and 1707 DF,  p-value: < 2.2e-16
```



$$\widehat{\text{price}} = 28,852 - 11,926 (\text{bedrooms}) + 125.8 (\text{sqft})$$

Alone, a bedroom was worth **+\$53,048**.

Next to square footage, it is worth **-\$11,926**.

The bedroom coefficient went negative



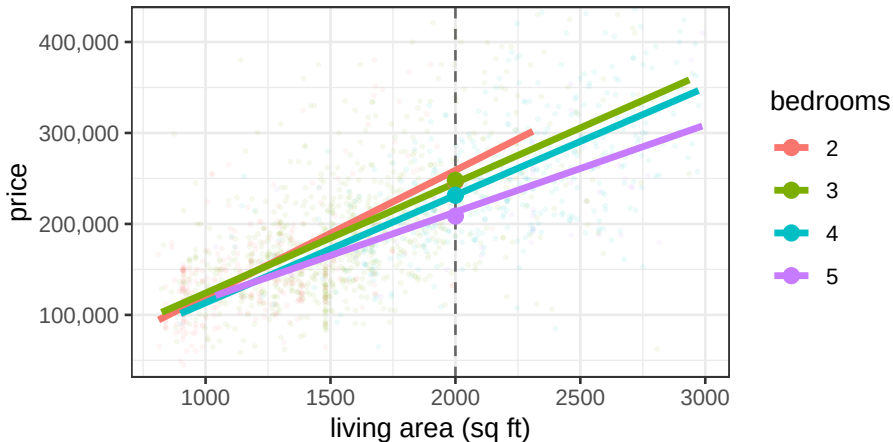
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Same houses. Same data. What changed?

Look at one bedroom count at a time



Four groups of houses. Within each one, bedrooms is **fixed**. Now slide out to 2,000 sq ft.



- The lines are **roughly parallel**. That common slope is $\hat{\beta}_2 = \$126$ per sq ft: what one more square foot buys you **without changing the bedroom count**.
- The lines are **stacked**, and more bedrooms sits **lower**. That vertical gap is $\hat{\beta}_1 = -\$11,926$: what one more bedroom does **at the same square footage**.

At 2,000 sq ft: a 3-bedroom is predicted at \$247,985, a 5-bedroom at \$208,619.



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Every multiple regression slope is read inside one line, never across them.

So why was the simple slope positive?



- Compare a 2-bedroom to a 5-bedroom house in the raw data and you are comparing a *small* house to a *big* one. You are sliding along the whole picture, not along one line.
- With size left out, bedrooms gets credit for the square footage it came with.
- Put size in the model and bedrooms has to answer a harder question: **take a house of a fixed size and cut it into more rooms**. Buyers pay less for that.
- Both slopes are correct. They answer **different questions**.

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Predictors	R^2	s_e
bedrooms	0.172	\$89,524
livingArea	0.515	\$68,552
both	0.520	\$68,219

- R^2 always rises when you add a variable, and the values do not add: $0.172 + 0.515 = 0.687$, but together they give only 0.520. The predictors overlap.
- $\sqrt{R^2} = 0.721$ is exactly the correlation between $\hat{Y}$ and Y .
- s_e is the typical miss on one house. R also prints adjusted R^2 , which penalizes extra parameters.



- **Overall null:** $\beta_1 = \beta_2 = 0$, equivalently $R^2 = 0$ in the population, equivalently the model beats guessing $\bar{Y}$ by nothing. Test it first with the F -test: here $F = 923$, $p < 2.2 \times 10^{-16}$. Rejected.
- **Individual coefficients:** test $H_0 : \beta_i = 0$ one at a time. This asks whether X_i adds anything **once the others are in**. Both of ours are significant.
- A smaller p -value does **not** mean a stronger predictor.



- Adding bedrooms is highly significant ($p = 2.7 \times 10^{-5}$), yet R^2 moves only from 0.515 to 0.520 and s_e barely budges.
- As a **prediction** tool it earns almost nothing.
- But it changed how we read the size coefficient entirely. Sometimes a variable is in the model not to predict, but **to hold something fixed**.
- **There is no universal rule for adding or removing variables.**

A slope is never just a number.

It is an answer to the question
“compared to what?”